Chapter 5 — Solids and Fluid Dynamics

Board-style MCQs · PECTAA Physics 11 · Answer key at end

Practice MCQs

1. Which solid has a regular repeating lattice?

(a) liquid **(b)** amorphous **(c)** polymeric **(d)** crystalline

2. Which solid softens over a range of temperature?

(a) metallic **(b)** ionic **(c)** crystalline **(d)** amorphous

3. Which is a polymeric solid?

(a) copper **(b)** wool **(c)** sodium chloride **(d)** glass

4. Stress is:

(a) extension/length **(b)** area/force **(c)** force/area **(d)** force × area

5. Strain is:

(a) force/area **(b)** area/force **(c)** extension/original length **(d)** length/extension

6. Young’s modulus is:

(a) extension × length **(b)** strain/stress **(c)** stress/strain **(d)** force × area

7. SI unit of stress is:

(a) newton **(b)** pascal **(c)** no unit **(d)** joule

8. Strain has:

(a) pascal **(b)** metre **(c)** newton **(d)** no unit

9. The region obeying Hooke’s law is up to:

(a) fracture point **(b)** proportional limit **(c)** ultimate strength **(d)** yield point only

10. Which is more elastic?

(a) wood **(b)** sponge **(c)** rubber **(d)** steel

11. Permanent deformation occurs beyond:

(a) proportional limit only **(b)** elastic limit **(c)** zero stress **(d)** origin

12. The maximum stress before fracture is:

(a) elastic limit **(b)** yield point **(c)** ultimate tensile strength **(d)** proportional limit

13. Energy stored in a stretched wire is:

(a) $F/\Delta L$ **(b)** $\frac{1}{2}F\Delta L$ **(c)** $\frac{1}{2}F/\Delta L$ **(d)** $F\Delta L$

14. The area under a force–extension graph gives:

(a) strain energy **(b)** power **(c)** Young’s modulus **(d)** stress

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15. Archimedes' principle gives upthrust equal to:

(a) density of body **(b)** weight of body **(c)** weight of displaced fluid **(d)** volume of body

16. Upthrust equals:

(a) mg **(b)** gV **(c)** ρV **(d)** ρgV

17. A body floats when upthrust is:

(a) greater than weight always **(b)** less than weight **(c)** zero **(d)** equal to weight

18. A steel ship floats because:

(a) steel is less dense than water **(b)** it has no weight **(c)** water has no viscosity **(d)** it displaces large volume of water

19. An ideal fluid is:

(a) opaque **(b)** turbulent **(c)** incompressible and non-viscous **(d)** viscous and compressible

20. The equation of continuity is:

(a) $U = \rho gV$ **(b)** $A_1v_1 = A_2v_2$ **(c)** $F = 6\pi\eta rv$ **(d)** $P + \frac{1}{2}\rho v^2 = \text{constant}$

21. The equation of continuity expresses conservation of:

(a) mass **(b)** charge **(c)** momentum **(d)** energy

22. When pipe area decreases, fluid speed:

(a) remains same **(b)** increases **(c)** decreases **(d)** becomes zero

23. Bernoulli's equation is:

(a) Y = stress/strain **(b)** $F = 6\pi\eta rv$ **(c)** $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$ **(d)** $A_1v_1 = A_2v_2$

24. In horizontal flow, where speed is high, pressure is:

(a) same **(b)** low **(c)** high **(d)** zero

25. The lift on an aeroplane wing is due to:

(a) density only **(b)** pressure difference **(c)** gravity **(d)** viscosity

26. A swinging ball curves due to:

(a) gravity only **(b)** pressure difference **(c)** electric force **(d)** magnetic force

27. Torricelli's theorem gives efflux speed:

(a) $\sqrt{(gh)}$ **(b)** gh **(c)** $\sqrt{(2gh)}$ **(d)** 2gh

28. Stokes' law is:

(a) F = kx **(b)** $F = 6\pi\eta rv$ **(c)** $F = \rho gV$ **(d)** F = ma

29. Terminal velocity is reached when:

(a) net force is zero **(b)** velocity is zero **(c)** drag is zero **(d)** weight is zero

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30. At terminal velocity, acceleration is:

(a) negative **(b)** g **(c)** maximum **(d)** zero

31. Terminal velocity of a sphere is proportional to:

(a) $1/r^2$ **(b)** r^2 **(c)** r **(d)** $1/r$

32. A superfluid has:

(a) zero viscosity **(b)** zero temperature **(c)** infinite density **(d)** infinite thermal conductivity

33. Superfluidity is observed in liquid helium below about:

(a) 100 K **(b)** 2 K **(c)** 273 K **(d)** 0°C

34. Cohesive force is between:

(a) same molecules **(b)** only gases **(c)** only solids **(d)** different molecules

35. Viscosity is:

(a) upthrust **(b)** internal friction of fluid **(c)** pressure **(d)** density

36. Streamline flow is:

(a) steady and orderly **(b)** always compressible **(c)** chaotic **(d)** always fast

37. Turbulent flow is:

(a) steady **(b)** irregular and mixing **(c)** incompressible only **(d)** non-viscous only

38. Pressure in a fluid increases with:

(a) area only **(b)** depth **(c)** temperature only **(d)** speed only

39. A Venturi mask works on:

(a) Pascal's law **(b)** Newton's third law **(c)** Bernoulli's principle **(d)** Archimedes' principle

40. Standing near a fast train is dangerous because:

(a) air speed lowers pressure **(b)** viscosity disappears **(c)** gravity increases **(d)** air speed raises pressure

41. Pure water spreads on glass because:

(a) cohesion exceeds adhesion **(b)** viscosity is zero **(c)** adhesion exceeds cohesion **(d)** density is zero

42. Mercury forms globules on glass because:

(a) adhesion exceeds cohesion **(b)** cohesion exceeds adhesion **(c)** it has no surface tension **(d)** it is ideal

43. In a real fluid, pressure along a uniform pipe decreases because of:

(a) viscosity **(b)** Bernoulli only **(c)** continuity **(d)** gravity only

44. Which fluid really exists in nature?

(a) superfluid only **(b)** ideal fluid **(c)** real fluid **(d)** perfect gas only

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45. The study of superfluids helps:

(a) only magnetism **(b)** only electricity **(c)** low temperature physics **(d)** only optics

46. Flow rate equals:

(a) A/v **(b)** Av **(c)** A + v **(d)** v/A

47. Young's modulus depends on:

(a) length only **(b)** material **(c)** load only **(d)** area only

48. In Searle's apparatus, the reference wire removes errors due to:

(a) area only **(b)** density only **(c)** load only **(d)** support yielding and temperature

49. A ductile material shows:

(a) no elastic region **(b)** large plastic deformation **(c)** brittle fracture only **(d)** no deformation

50. The yield point is where:

(a) fracture occurs **(b)** stress is zero **(c)** plastic deformation begins **(d)** Hooke's law starts

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Answer Key with Brief Reasons

1. Answer: (d) crystalline — Crystalline solids have long-range order.

2. Answer: (d) amorphous — Amorphous solids lack a sharp melting point.

3. Answer: (b) wool — Wool is a natural polymer.

4. Answer: (c) force/area — $\sigma = F/A$.

5. Answer: (c) extension/original length — $\varepsilon = \Delta L/L_0$.

6. Answer: (c) stress/strain — $Y = \sigma/\varepsilon$.

7. Answer: (b) pascal — Stress = N/m^2 = Pa.

8. Answer: (d) no unit — Strain is a ratio of lengths.

9. Answer: (b) proportional limit — Stress ∝ strain up to proportional limit.

10. Answer: (d) steel — Steel has larger Young's modulus.

11. Answer: (b) elastic limit — Plastic deformation begins after elastic limit.

12. Answer: (c) ultimate tensile strength — UTS is the peak stress.

13. Answer: (b) ½FΔL — Average force ½F times extension.

14. Answer: (a) strain energy — Area = work stored.

15. Answer: (c) weight of displaced fluid — U = weight of displaced fluid.

16. Answer: (d) ρgV — U = ρ_fluid g V_displaced.

17. Answer: (d) equal to weight — Equilibrium requires U = W.

18. Answer: (d) it displaces large volume of water — Hollow shape gives enough upthrust.

19. Answer: (c) incompressible and non-viscous — Ideal flow assumes no viscosity and constant density.

20. Answer: (b) $A_1v_1 = A_2v_2$ — Mass conservation for incompressible flow.

21. Answer: (a) mass — Mass flow rate is constant.

22. Answer: (b) increases — Av = constant.

23. Answer: (c) $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$ — Energy conservation along streamline.

24. Answer: (b) low — $P + \frac{1}{2}\rho v^2 = \text{constant}$.

25. Answer: (b) pressure difference — Faster air above gives lower pressure.

26. Answer: (b) pressure difference — Spin creates speed difference.

27. Answer: (c) $\sqrt{(2gh)}$ — $v = \sqrt{(2gh)}$.

28. Answer: (b) $F = 6\pi\eta rv$ — Viscous drag on a sphere.

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29. Answer: (a) net force is zero — Drag + upthrust = weight.

30. Answer: (d) zero — Net force is zero.

31. Answer: (b) r² — $v_t = 2gr^2\rho/9\eta$.

32. Answer: (a) zero viscosity — Superfluids flow without internal friction.

33. Answer: (b) 2 K — Helium-4 becomes superfluid near 2 K.

34. Answer: (a) same molecules — Cohesion is like-molecule attraction.

35. Answer: (b) internal friction of fluid — Viscosity opposes relative motion of layers.

36. Answer: (a) steady and orderly — Layers slide smoothly.

37. Answer: (b) irregular and mixing — Eddies and mixing occur.

38. Answer: (b) depth — $P = \rho gh$.

39. Answer: (c) Bernoulli's principle — Fast oxygen flow draws air.

40. Answer: (a) air speed lowers pressure — Pressure difference can pull a person.

41. Answer: (c) adhesion exceeds cohesion — Water wets glass.

42. Answer: (b) cohesion exceeds adhesion — Mercury does not wet glass.

43. Answer: (a) viscosity — Viscous losses reduce pressure.

44. Answer: (c) real fluid — All natural fluids have some viscosity.

45. Answer: (c) low temperature physics — It reveals quantum fluid behaviour.

46. Answer: (b) Av — Volume per second = area × speed.

47. Answer: (b) material — It is a material property.

48. Answer: (d) support yielding and temperature — Both wires share the same support.

49. Answer: (b) large plastic deformation — It can be drawn into wires.

50. Answer: (c) plastic deformation begins — Permanent strain begins.

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